

Triplet Analyzing Unit

Methodology, Benchmarking, and Insights from Low-Similarity Knowledge Graph Pairs

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1. Context and Purpose

In our evaluation pipeline, each instance is represented as one or more knowledge graphs (KGs) constructed from structured semantic triplets (Subject, Relation, Object). We first compute similarity between KG pairs. The lowest-scoring 5% of instances are then routed to a Triplet Analyzing Unit for deeper diagnosis. The goal of this unit is to move from a single similarity score to an interpretable explanation of which facts aligned and what types of factual differences remain.

2. Triplet Analyzing Unit Overview

Given two KGs, the unit produces two outputs:

- Aligned triplet pairs: meaning-equivalent facts matched across the two graphs.
- Residual diagnostics: a structured breakdown of non-aligned facts, categorized into interpretable error classes.

3. Methodology: Step-by-Step Flow

The Triplet Analyzing Unit follows a six-stage flow. The diagram below summarizes the process.

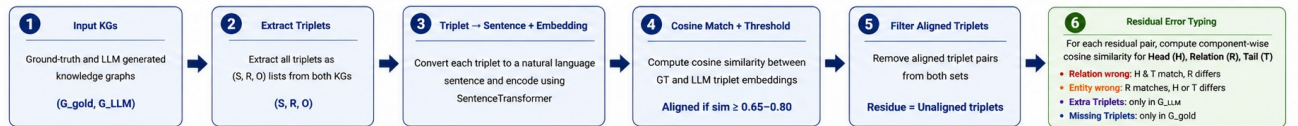


Figure 1: Triplet Analyzing Unit flow (six-stage alignment and residual typing).

Step details:

1. Input KGs: take the two graphs to be compared (e.g., reference KG and generated KG).
2. Extract triplets: represent each graph as a list of (S, R, O) triplets.
3. Triplet to sentence + embedding: convert each triplet into a short sentence-like text and encode using a SentenceTransformer model.
4. Cosine matching + threshold: compute cosine similarity between sentence embeddings; treat a pair as aligned if similarity exceeds a chosen threshold (typically 0.65 to 0.80 depending on strictness).
5. Filter aligned triplets: remove aligned pairs from both sets; the remainder forms the residual (non-aligned) triplets.

6. Residual error typing: for residual pairs, compute component-wise cosine similarity on Head (H), Relation (R), Tail (T) to assign interpretable mismatch classes.

4. Residual Error Typing Rules

After filtering aligned triplets, the unit categorizes the remaining differences using component-wise cosine similarities (H/R/T). A common threshold (e.g., 0.70) is applied to decide whether each component matches.

- Relation wrong: head and tail match (entities consistent) but relation diverges.
- Entity wrong: relation matches but the entity pair is inconsistent (subject and/or object differs).
- Extra triplets: triplets present only in the generated KG (unsupported additions / hallucinated facts).
- Missing triplets: triplets present only in the reference KG (omitted facts).

5. Benchmarking Setup

To evaluate alignment quality, we benchmarked our method against the KEA baseline on manually annotated subsets of health-domain datasets. The evaluation focuses on aligned-triplet detection, using gold aligned triplet pairs as the reference.

5.1 Manual Annotation and Agreement

Aligned triplet pairs were labeled by three medical-student annotators. Agreement was computed over the candidate set of aligned pairs (union of all pairs proposed by at least one annotator). The agreement results show strong consistency and support the reliability of the gold labels.

Agreement Metric	Value
Percent agreement (all 3 agree)	0.9433
Pairwise F1 (range)	0.9708 – 0.9882
Pairwise Jaccard (range)	0.9433 – 0.9766

Table 1: Inter-annotator agreement for aligned-triplet labeling.

5.2 Evaluation Metrics

We report Precision, Recall, and F1 for aligned-triplet detection using both micro and macro averaging. Micro aggregates counts over all triplets across the dataset, while macro averages performance per instance and then averages across instances, reflecting consistency across questions.

5.3 KEA Baseline vs Our Method

KEA performs embedding-based retrieval followed by component-wise similarity checks on subject, relation, and object. Our method uses sentence-level semantic similarity for alignment, which is more robust to relation paraphrases, and then provides residual error typing for interpretability.

Metric	KEA	Ours
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Micro F1	0.836	0.895
Macro F1	0.622	0.782

Table 2: Summary benchmarking (aligned-triplet detection) for KEA vs our method.

For completeness, the benchmarking summary used in the poster is shown below.

KEA Baseline	0.518	0.649	0.576
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Triplet Unit Benchmarking: Gold triplets labeled by 3 medical-student annotators.

Metric	KEA	Ours	Agreement	Value
Micro F1	0.836	0.895	All-3 agree	0.9433
Macro F1	0.622	0.782	Pairwise F1	0.9708–0.9882

Figure 2: Poster benchmarking summary (metrics + agreement).

6. Insights from the Lowest-Similarity 5% Cases

After global KG similarity scoring, the lowest-similarity 5% cases were routed to the Triplet Analyzing Unit. Across both MessaQA and PubMedQA and across the three models (Mistral, Llama, Gemma), the diagnostics support three recurring insights:

- Model comparison on hard cases: Mistral remains the most reliable on the low-similarity set, recovering more reference-aligned triplets; Gemma is typically the weakest, particularly on MessaQA.
- Domain difficulty: PubMedQA hard cases show sharply lower aligned-triplet recovery across all models, consistent with biomedical terminology, abbreviations, and entity variability dominating failure modes.
- Failure characterization: in many low-similarity cases, the residual set contains substantial Extra and Missing triplets, indicating that failures often involve unsupported additions and/or omission of key reference facts rather than minor phrasing variation.

The insight summary used in the poster is shown below.

On the lowest 5% similarity pairs, Mistral-7B remains the most reliable, recovering more reference-aligned triplets than other models, while Gemma-7B is weakest, particularly on MesaQA. PubMedQA hard cases show sharply lower aligned-triplet recovery across all models, confirming that biomedical terminology and entity variability dominate failure modes.

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Figure 3: Poster insight statement for the lowest 5% similarity cases.

7. What the Unit Produces in Practice

For each analyzed instance, the unit produces (i) aligned triplet pairs with similarity scores and (ii) categorized residual triplets (relation wrong, entity wrong, extra, missing). These outputs can be aggregated across datasets and models to identify systematic patterns, such as common relation paraphrases that KEA misses, frequent entity grounding errors, and the prevalence of extra facts.

8. Limitations and Next Steps

- Threshold sensitivity: alignment performance depends on the chosen sentence-level similarity threshold; tuning trades precision for recall.
- Entity normalization: biomedical term variability can reduce component-level matching; improved canonicalization and abbreviation handling should improve residual typing.
- Broader evaluation: ongoing annotation across additional datasets and multiple LLMs will strengthen the conclusions and test robustness.